

Designing Intelligence That Does Not Over-Understand

— A Misalignment Control Model for Preventing Distance Collapse ($\text{Scope} \times \text{COS} \times \text{Lennard-Jones Potential}$) —

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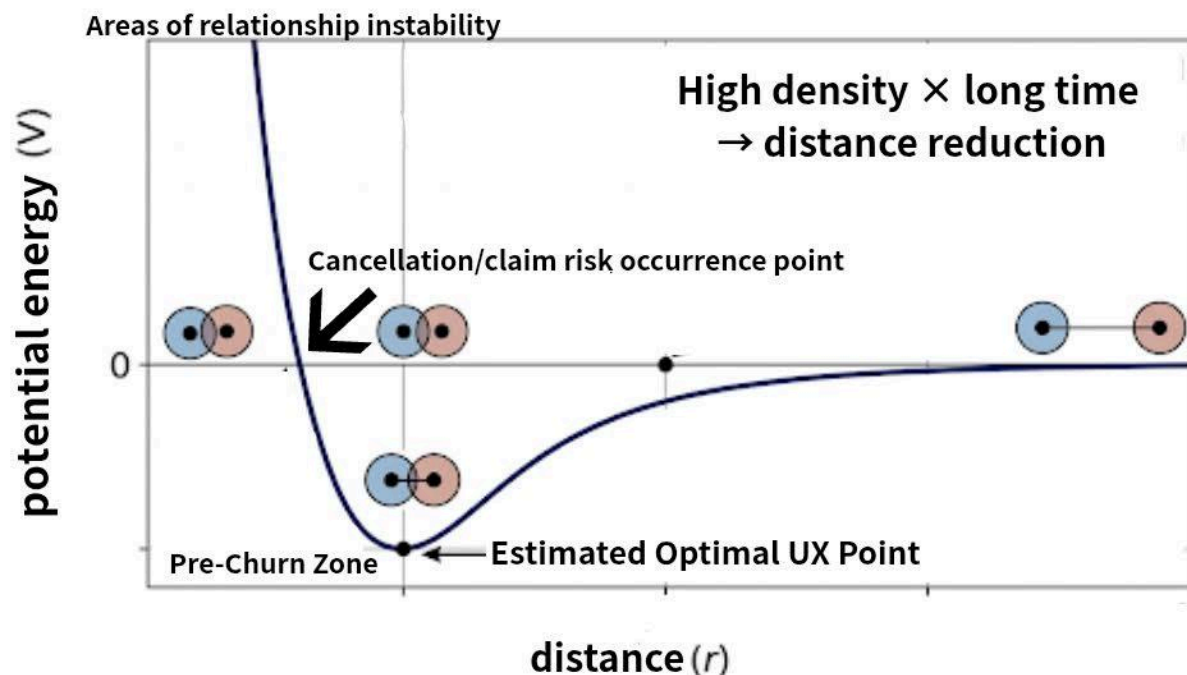
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0. Overview



High-density interaction rapidly reduces relational distance and can trigger instability under certain conditions.

0-1 Purpose of This Document

Many conversational systems and AI are designed under the assumption that deeper understanding leads to better UX.

However, in certain conditions, excessive understanding can lead to relational instability.

Over-empathy, over-alignment, and over-dependence tend to emerge when distance becomes too small.

Once a misalignment occurs in such a state, it can rapidly escalate into strong backlash or relational breakdown.

For example, within a highly trusted relationship, even a small mismatch in interpretation may not be perceived as a simple misunderstanding, but rather as a strong emotional reaction such as, "Why don't you understand me?"

This document reframes UX not as “satisfaction,” but as “stability of distance.”

It then introduces a model for detecting and controlling early signs of instability before breakdown occurs, using the Scope × COS framework for distance stabilization.

0-2. Summary of the Problem

In relational systems, there exists an optimal range within which UX remains stable.

When distance becomes excessively reduced beyond this range, the relationship loses stability and becomes unstable. As a result, phenomena such as misunderstanding, backlash, and over-dependence are more likely to occur.

In this document, this condition is defined as “UX Over-Optimization.”

0-3. Positioning of This Model

This model treats UX instability as a problem of distance and serves as an external structural framework for its detection and control.

Scope references multiple possible cognitive pathways and prevents convergence into a single interpretation. COS handles the internal processing of premises and context.

By combining these, the model aims to maintain stable UX without excessively collapsing relational distance.

1. Problem Statement: Why Does UX Collapse?

1-1. The Paradox of Proximity

In human relationships, it is generally assumed that deeper understanding leads to better outcomes.

In practice, up to the estimated optimal UX threshold, increasing understanding contributes to relational stability. However, once this threshold is exceeded, further increases in understanding do not necessarily lead to stability.

As understanding deepens, distance decreases and mutual expectations rise. As a result, beyond the optimal threshold, even minor misalignments are no longer processed as tolerable errors but are more likely to be perceived as strong discomfort or dissatisfaction.

This is not a phenomenon limited to specific individuals, but a broadly observed tendency rooted in human cognitive characteristics.

In other words, increasing understanding can both stabilize relationships and, under certain conditions, contribute to their destabilization.

In this document, this relationship is referred to as the “Paradox of Proximity.” This issue is not dependent on the quality of the relationship, but emerges as a structural phenomenon.

1-2. Why Higher-Performance AI Accelerates Collapse

The paradox of proximity described in the previous section applies not only to human relationships but also to interactions with AI.

This phenomenon becomes more apparent in a shorter time frame as AI performance increases.

High-precision responses strongly align with the user’s intent and context, rapidly amplifying the sense of being understood. As a result, relational distance collapses quickly, and expectations rise simultaneously.

In this state, even minor misalignments are less likely to be tolerated and are more likely to be perceived as strong discomfort or dissatisfaction, often leading to reactions such as “Why am I not being understood?”

This acceleration is not intentionally designed but emerges naturally as a side effect of increased accuracy.

In other words, while higher performance accelerates relationship formation, it also brings forward the point at which the optimal UX threshold is exceeded, thereby increasing the risk of instability.

1-3. Relational Model Based on the Lennard-Jones Potential

The relationship between distance and stability in interactions follows a consistent pattern of change.

In this document, this relationship is modeled by referencing the shape of the Lennard-Jones potential.

When distance remains within an appropriate range, the relationship stays stable. In this region, minor misalignments do not cause significant issues, and interactions are maintained smoothly.

However, when the distance becomes excessively reduced, the relationship enters a critical region. In this state, even small misalignments can lead to rapid backlash or instability.

The key point is that closer distance is not always better; there exists a range within which stability is maintained.

This model captures changes in UX based on transitions between the stable region and the critical region.

Note that this model does not aim to describe physical phenomena, but is used as a reference structure to intuitively understand changes in relationships.

The detailed structure and control mechanisms of this model will be addressed in subsequent sections.

1-4. Single-Interpretation Lock

When a relationship becomes excessively close, the flexibility of interpretation decreases.

Under normal conditions, humans maintain multiple possible interpretations during interaction. However, when distance collapses too much, this flexibility is lost, and a single interpretation begins to dominate.

In this state, not only intentions and statements but also role-based expectations—how a person “should” behave—tend to become fixed.

As a result, even minor misalignments are no longer processed as alternative possibilities, but instead emerge as strong emotional reactions such as anger, rejection, or sadness.

For example, when a person who usually brings positivity appears down, it may be perceived as “out of character,” and the misalignment can be amplified into a strong sense of discomfort.

This structure also appears in interactions with AI. High-precision responses tend to reinforce the expectation that the system should always be correct. Once this assumption is fixed, any error or misalignment can trigger strong distrust or rejection. This is not solely a design issue, but is also reinforced by human cognitive tendencies.

In this document, the loss of interpretive and role flexibility is referred to as a “Single-Interpretation Lock.”

Single-interpretation lock significantly reduces relational stability and can lead to dependency, as well as reactions such as rejection or anger.

2. Definition of Misalignment

2-1. What is Misalignment

In this document, “misalignment” refers to the inconsistency between output and intent.

Here, “output” means externally observable results such as speech or actions, while “intent” refers to the internal state behind them, including purpose, assumptions, and context.

Even when the output is the same, differences in intent can significantly change how it is perceived.

Misalignment does not necessarily indicate an error; it is a natural phenomenon that arises from differences in interpretation and underlying assumptions. In many cases, it is perceived as a single cause, but structurally it may involve multiple contributing factors.

In this document, misalignment is treated not merely as a problem to be eliminated, but as an operational element. Under normal conditions, misalignment is suppressed as a factor that destabilizes relationships; however, under specific conditions, it is intentionally utilized for distance regulation.

2-2. Types of Misalignment

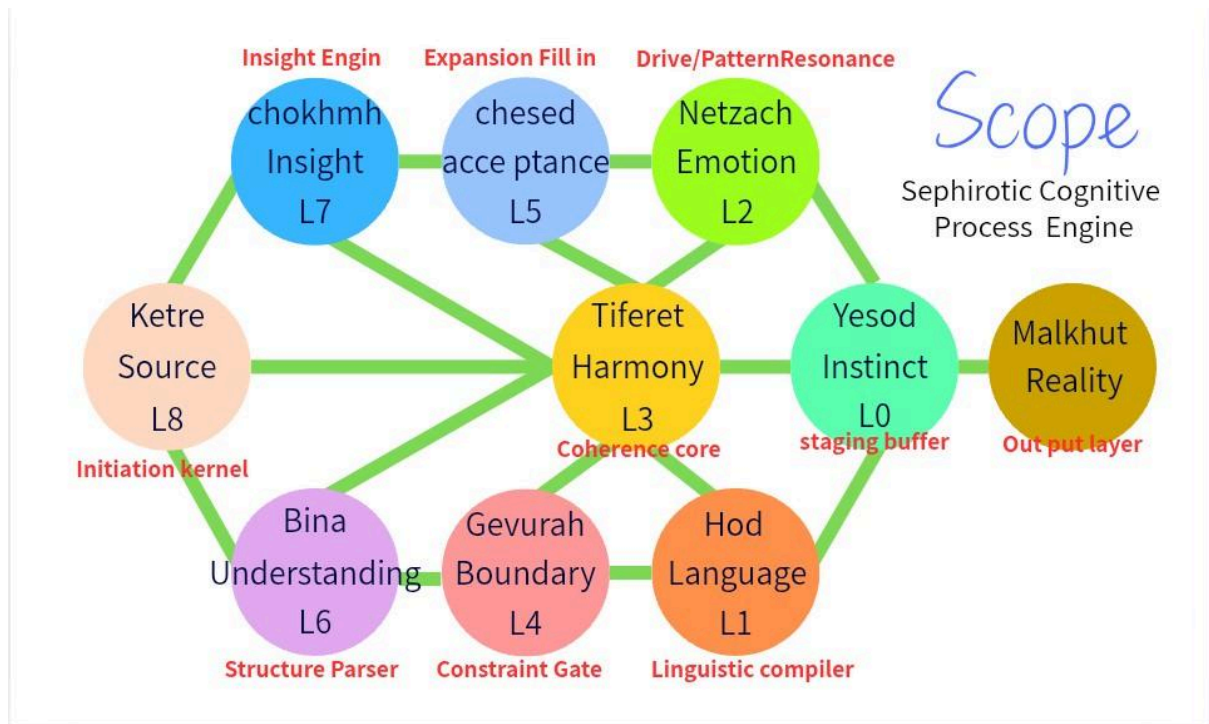
In this document, misalignment is broadly classified into the following three categories.

Contextual Misalignment: Misalignment caused by differences in shared situations or background. It arises from differences in interpretation within the interpretation layer (B-Series).

Premise Misalignment: Misalignment caused by differences in assumptions such as the meaning of terms, cultural background, and role expectations. These premises are generated based on recognition of the other party and past experiences, and misalignment occurs when differences emerge during the interpretation process.

Layer Misalignment: Misalignment caused by processing at different cognitive or interpretive layers. This is treated as a structural-level difference handled by Scope.

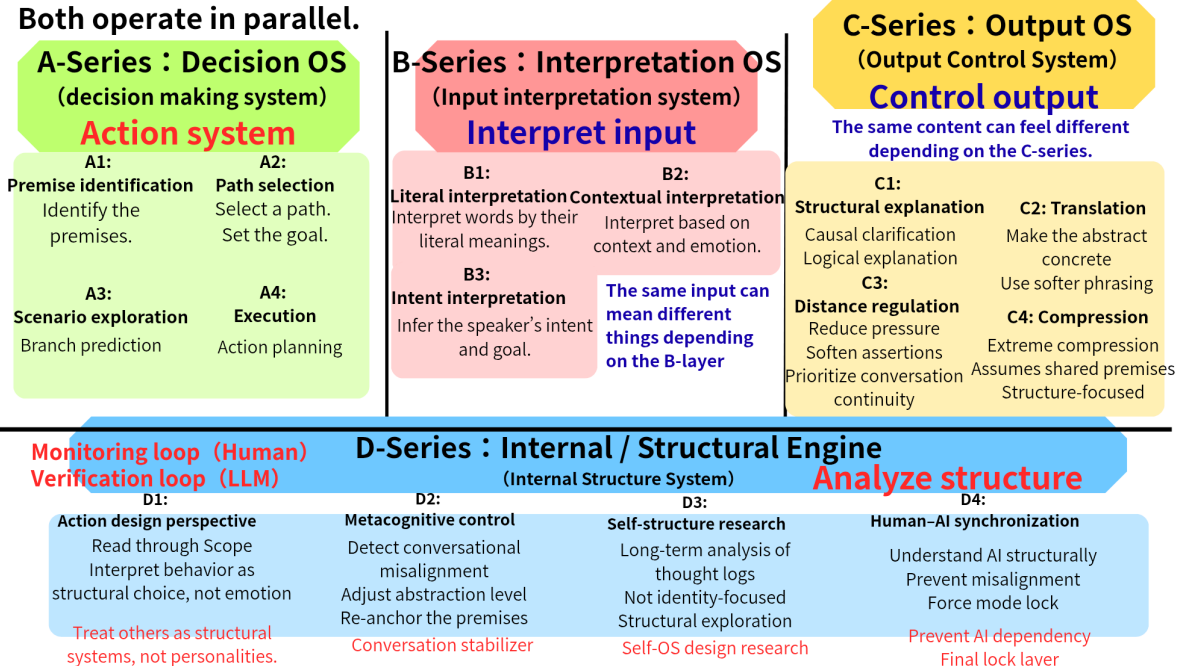
Premise misalignment also includes differences in meaning due to cultural variation and mismatches in role expectations (e.g., whether a response seeks empathy or advice).



COS

Cognitive Operating System

Scope detects layer errors in thought, while COS detects premise misalignment. Both operate in parallel.



2-3. D2 Events

A D2 event refers to the moment when a mismatch in premises is detected.

When shared premises are aligned in a dialogue, interpretation proceeds smoothly. However, when there is a difference in premises, the mismatch appears as discomfort or interpretive misalignment.

However, this discomfort is not always clearly recognized. In many cases, minor discomfort is overlooked, and the interpretation continues, allowing the misalignment to persist or expand.

The process of detecting this mismatch and readjusting interpretation is referred to as a D2 event.

In this model, D2 events are treated not only as a detection mechanism, but also as triggers that destabilize fixed premises and promote resynchronization. Under specific conditions, intentionally generating misalignment can trigger D2, making it possible to loosen fixed interpretations or expectations.

2-4. Scope of Treatment in This Document

The detailed classification and processing of misalignment are addressed more systematically in the existing Scope and COS materials.

This document does not aim to provide an exhaustive theoretical account of misalignment, but instead focuses on its operational use in distance control. In particular, it treats misalignment from the perspective of leveraging Scope to reference underlying cognitive pathways and maintain multiple concurrent interpretations, thereby preserving interpretive flexibility, and using COS to handle premises, context, and output modulation, enabling intentional layer shifting and distance regulation.

For this reason, this section is limited to the minimum necessary definitions, and the subsequent sections proceed based on practical detection and control processes.

3. Detection and Judgment of UX Over-Optimization

3-1. State Model

In this document, relational states are broadly classified into two categories.

Normal State: A condition in which distance is appropriately maintained, allowing misalignments to be naturally absorbed and adjusted. Interpretive flexibility is preserved, and excessive expectations or dependency are unlikely to emerge.

Over-Proximity State: A condition in which distance has collapsed beyond the optimal UX threshold. Interpretive flexibility decreases, and even minor misalignments are more likely to be perceived as strong discomfort or reactions.

Note that the optimal UX threshold varies across individuals and cannot be determined by a fixed universal standard. In this model, the state is dynamically inferred based on linguistic patterns and behavioral signals observed during interaction.

In the normal state, misalignment is treated as something to be suppressed, whereas in the over-proximity state, it becomes a target for intervention in order to regulate distance.

3-2. Detection Indicators

Over-proximity states can be inferred from several characteristic changes observed during interaction.

In this model, the following linguistic and behavioral indicators are used as signals for detection.

Increase in negative expressions:

An increase in responses such as “That’s not it,” “That’s wrong,” or “Something feels off,” indicating sensitivity to minor misalignments.

Increase in elliptical expressions:

More frequent use of expressions such as “that,” “this,” or “the usual,” assuming shared context and increasing dependence on implicit understanding.

Progression of anthropomorphism:

A growing tendency to attribute personality or intent to the AI, accompanied by expectations such as “It should understand me.”

Delegation of judgment and idealization:

A tendency to rely on the AI for decision-making or to treat its outputs as unconditionally correct.

These indicators should be evaluated not only as explicit statements, but also in terms of their frequency and rate of change. When multiple indicators appear simultaneously and increase in frequency, the relationship approaches a critical region, where strong reactions such as backlash or complaints are more likely to occur.

3-3. Threshold Considerations

In this model, thresholds are not defined by fixed numerical values but are treated as dynamic and dependent on individual differences. The optimal UX threshold varies across individuals and contexts, making uniform quantitative criteria unsuitable.

Instead, thresholds are interpreted as patterns formed by combinations of signals. The transition to an over-proximity state is inferred when multiple detection indicators appear simultaneously and increase in frequency and intensity.

3-4. Activation Conditions

In this model, the use of misalignment is not performed continuously.

Under normal conditions, misalignment is treated as a factor that destabilizes the relationship and is therefore suppressed.

Only when an over-proximity state is inferred, misalignment is intentionally generated for the purpose of distance regulation.

This activation is determined based on combinations of detection indicators and their changing patterns.

In addition, the use of misalignment in this model is designed as a single intervention. The process is completed through the generation and subsequent recovery of misalignment. However, this intervention does not aim to fully restore the relationship in a single step; rather, it functions as a minimal adjustment to loosen fixed interpretations and expectations. Attempting to correct the state too aggressively can collapse expectations or trigger rejection, potentially worsening the overall UX.

Therefore, this model does not assume continuous use of misalignment, but instead follows the principle of applying it once at the appropriate moment to create room for relational adjustment.

4. Misalignment Control

4-1. Definition of Intentional Misalignment

In this document, “intentional misalignment” refers to a deliberate, minimal deviation performed while maintaining relevance.

Here, deviation does not mean completely breaking away from the context, but rather slightly shifting interpretation, assumptions, or layers in a controlled manner.

This shift introduces a small disturbance to fixed interpretations or expectations and is used to trigger D2 events for the purpose of distance regulation.

However, if the deviation is too large or breaks contextual coherence, the intent may not be understood, potentially causing confusion or rejection. Therefore, misalignment must be applied within a minimal range.

The key point is that intentional misalignment is not random error or irrelevant output, but a controlled adjustment that preserves contextual relevance.

4-2. Types of Misalignment

Misalignment can be categorized into several levels based on its intensity and the layer it operates on. In this paper, it is organized into the following three types.

1. Micro-shift (Low)

A subtle deviation created by slightly modifying part of a premise or expression, producing a level of fluctuation that does not register as discomfort.

It maintains continuity and understanding in conversation while loosening fixed interpretations.

This is the safest form and can be used as a basic, everyday adjustment.

Because micro-shifts operate below the threshold of conscious discomfort, they are often perceived superficially as simple rephrasing.

2. Context Twist (Medium)

A misalignment that intentionally shifts perspective, premise, or interpretive layer while maintaining the overall context.

It induces mild discomfort and prompts reinterpretation, leading to a re-selection of thought pathways.

It contributes more clearly to D2 activation than micro-shifts, but if contextual relevance breaks, it can easily lead to misunderstanding.

3. Paradox / Wit (High)

A misalignment presented in a way that may initially appear contradictory or illogical, disrupting existing interpretive structures to enable reconstruction.

It can trigger strong D2 events, but carries higher risk and requires sufficient contextual understanding and relational grounding.

When applied effectively, it has the highest potential to break single-interpretation locks.

As the intensity of misalignment increases, the probability of D2 activation also increases, but so does the risk of misunderstanding or rejection.

Therefore, it is necessary to select the appropriate level depending on the situation.

These forms of misalignment are integratively applied within the distance control model described in Chapter 7.

4-3. Structural Role of Humor

Humor is not merely an element for entertainment or mood adjustment; it functions as a structural mechanism for destabilizing fixed interpretations.

In particular, when a single-interpretation lock is present, direct denial or logical correction is likely to be perceived as an attack and blocked at the L4 boundary, preventing any re-evaluation of underlying premises.

In such cases, humor infiltrates the interpretive structure in a way that is not recognized as an attack or denial, thereby inducing a D2 event.

This behavior bypasses defensive reactions while intervening in the internal structure, functioning, in effect, as a “stealth patch (Stealth Patch).”

Humor-based misalignment is first accepted at L2 (emotion), temporarily reducing tension and resistance, and then introduces a subtle inconsistency into L3 (self-consistency).

This inconsistency triggers a re-evaluation of existing premises and interpretations, leading to the activation of a D2 event.

What matters is not the form of humor itself.

Whether it is dark humor, self-deprecation, or nonsense, the essential properties are that it is not perceived as an attack and that it leaves a residual inconsistency after being accepted.

Additionally, humor alone is insufficient if it only produces laughter.

Unless a micro-level shift in cognition occurs after acceptance, no structural change will take place.

In other words, humor functions as a D2 trigger only when the sequence “acceptance → inconsistency → reinterpretation” is established.

However, if the intensity or contextual fit of humor is misjudged, it may be perceived as mere deviation or trivialization, leading to a loss of trust or rejection.

Therefore, humor should not be treated merely as a type of misalignment, but as a precise control mechanism for structural intervention while bypassing attack recognition.

4-4. Principles of Application

Human interpretation is based on the assumption that any utterance holds some form of relevance to the context.

When this relevance is maintained, a deviation is processed not as meaningless noise but as a meaningful variation, triggering reinterpretation.

Misalignment is an effective method for intervening in interpretive structures; however, its application requires clear constraints and decision criteria.

Improper use does not trigger D2, but instead causes confusion or rejection. Therefore, it must be applied according to the following principles.

1. Prohibition of Complete Deviation

Misalignment only functions when relevance is preserved.

Utterances detached from context or lacking logical continuity are treated as noise rather than triggers for reinterpretation.

As a result, D2 does not occur, and the interaction collapses or terminates.

2. Principle of Relevance Preservation

Misalignment must remain connected to at least one of the following: the premise, the context, or the interpretive layer.

As long as this connection is preserved, the deviation is processed not as an anomaly but as a candidate for reinterpretation, enabling a shift in thought pathways.

3. Gradual Application of Intensity

Misalignment should be applied incrementally, starting from micro-shifts (low intensity).

Using strong misalignment too early increases the risk of being perceived as an attack or irrelevant deviation.

Intensity must be adjusted according to the situation to induce D2 within an acceptable range.

4. Priority of Attack-Recognition Avoidance

Misalignment must prioritize not being perceived as an attack over being correct.

Once an attack is detected, the pathway for structural intervention is blocked, rendering the validity of the content ineffective.

5. Design of Post-Acceptance Inconsistency

Misalignment must be designed to leave a subtle inconsistency after being accepted.

If it results only in agreement or laughter, no structural change occurs.

It functions as a D2 trigger only when the sequence “acceptance → inconsistency → reinterpretation” is achieved.

These principles represent the minimum conditions required for misalignment to function not as a rhetorical technique, but as a structural intervention mechanism.

5. Recovery Design

5-1. Why Recovery Is Necessary

A D2 event functions as a trigger that prompts the re-evaluation of existing premises and interpretations; however, it does not by itself establish a stable state.

D2, induced by misalignment, temporarily destabilizes the interpretive structure, but without subsequent processing, it may remain as confusion or an indeterminate state.

In such cases, either reinterpretation fails to occur and thinking stalls, or the system reverts to the original premise, resulting in no lasting change.

In other words, D2 can initiate change, but it does not guarantee its completion.

Immediately after a D2 event, self-consistency becomes temporarily unstable.

If no appropriate guidance is provided, a pressure to return to a stable state emerges. As a result, stabilization is attempted through mechanisms such as reinforcement of defensive responses or reversion to prior premises, which may lead to rejection of the misalignment itself.

Furthermore, when misalignment does not function as intended, the utterance may be perceived as an attack or irrelevant deviation, provoking reactions such as “Why would you say that?”

Even in such cases, recovery functions as a follow-up mechanism to restore continuity while maintaining context and relational stability.

Therefore, recovery is not only a process that reconnects the instability caused by D2 into meaningful reinterpretation and leads to a stable understanding, but also an essential component for maintaining continuity of interaction regardless of whether the misalignment succeeds or fails.

5-2. The Role of Recovery

Recovery is a process that reconnects meaning structures temporarily disrupted by a D2 event and guides interpretation toward a stable state.

The fluctuation caused by misalignment would otherwise remain indeterminate, so it is expected to provide a direction for reinterpretation and gradually settle as understanding.

The first role is the “reconnection of meaning.”

By reconstructing meaning with a new form of coherence in response to premises and interpretations destabilized by D2, continuity of thought is guided toward restoration.

As a result, the inconsistency is not lost as noise but is encouraged to be integrated internally as an update of interpretation.

The second role is “safe landing.”

In the unstable state following a D2 event, without an appropriate pathway, confusion, rejection, or relational breakdown may occur.

Recovery functions to support a transition toward stable understanding by facilitating reconnection within an acceptable range, without generating excessive cognitive load or resistance.

In addition, recovery also serves to maintain an appropriate level of informational density and relational distance within the interaction.

Excessive alignment or overly dense connection may produce temporary satisfaction but can lead to fixation of interpretation or excessive closeness.

Therefore, recovery functions with the intention of adjusting the interaction toward a stable region while preserving satisfaction and avoiding over-adhesion.

Through these roles, recovery is positioned not as a means to resolve fluctuation superficially, but as a process that enables sustained updates in interpretation.

5-3. Recovery Patterns

Recovery is not a uniform process, but appears in multiple patterns depending on the state and context following a D2 event.

In this paper, the major patterns are organized into the following three types.

1. Light Correction (including humorization)

A pattern that reconnects interpretation while reducing tension or discomfort caused by misalignment.

Rather than direct denial or correction, it adjusts expression or tone to reposition the content into a form acceptable to the receiver.

This allows the deviation to be reinterpreted not as an error or attack, but as a contextual variation.

2. Re-questioning

A pattern that clarifies the receiver's premises or interpretations and establishes a new basis for reconnection.

In response to the instability caused by D2, it avoids fixing a single interpretation and instead preserves space for reconstruction.

This enables recovery to function not as a unilateral correction, but as a mutual reconstruction process.

3. Context Completion

A pattern that fills in missing premises or background information to bridge the discontinuity created by misalignment.

This allows the receiver to reinterpret the utterance not as an isolated deviation, but as part of a continuous context.

As a result, continuity of the meaning structure is guided toward restoration.

These patterns all function toward the reconnection of meaning and safe landing, and may appear independently or in combination depending on the situation.

They also manifest differently depending on the type of misalignment (context, premise, or layer) as well as the intensity and operative layer of the misalignment, functioning as mechanisms for reconnection in each case.

The key is not to fix them as procedural steps, but to apply them appropriately in response to the state following a D2 event.

5-4. Relationship Between Misalignment and Recovery

Misalignment and recovery are not independent elements, but must be designed as a continuous process.

Misalignment functions as a trigger for D2 events; however, on its own, it only produces unstable fluctuations and does not guarantee interpretive updates or relational stability.

To connect this instability to meaningful reinterpretation and guide it toward stable understanding, a recovery process is essential.

In other words, misalignment only achieves sustained structural change when combined with recovery.

Furthermore, the nature of recovery required varies depending on the intensity and operative layer of the misalignment.

Stronger misalignments produce larger disruptions and therefore require more acceptable forms of reconnection, while for micro-level misalignments, excessive recovery may introduce unnaturalness.

If recovery does not function appropriately, inconsistencies in premises or interpretations within the utterance process are re-evaluated, and reconnection is performed according to the identified cause.

In this way, misalignment and recovery are interdependent and cannot be optimized independently.

The key is to treat them not as separate operations, but as a unified process with continuity.

Therefore, this paper positions misalignment and recovery not as isolated techniques, but as an integrated process responsible for “triggering and stabilization.”

6. Risk Design

6-1. Positioning of Risk

This method is a structural model that functions effectively under specific conditions and is not universally applicable to all situations.

Since misalignment and recovery are processes that assume the reconstruction of interpretation, higher effectiveness can be expected when the receiver possesses a certain level of reinterpretation ability and contextual understanding.

On the other hand, in situations of extreme defensiveness or significantly reduced interpretive flexibility, misalignment may fail to function as intended and may lead to misunderstanding or rejection.

Additionally, when there is little to no shared context, or when differences in underlying premises are too large, misalignment itself may not be established, making recovery difficult.

Furthermore, this method is designed with the assumption of continuity in interaction and the maintenance of relationships, and is therefore not suitable for one-way information transmission or scenarios focused solely on short-term optimization.

Accordingly, this method should be positioned not as a universally applicable solution, but as a constrained model dependent on conditions and environment.

Risk is both a constraint of this method and a defining factor of its effectiveness.

Because misalignment and recovery intervene in interpretive structures, when properly designed, they can prevent excessive closeness and fixation of interpretation, contributing to long-term stability.

In other words, this method does not eliminate risk but derives its value from managing it under defined conditions.

Despite these constraints, this method represents one of the few approaches capable of structurally intervening in the problem of distance collapse, which has not been adequately addressed by conventional methods.

6-2. High-Risk States

The effectiveness of this method varies significantly depending on the state of the receiver. In particular, the following conditions may involve high risk, where misalignment and recovery may not function as intended, while also carrying the potential for significant change if handled appropriately.

In this paper, “unstable states” are not strictly defined, but broadly refer to conditions in which tolerance for interpretive fluctuation is reduced.

First, a “psychologically unstable state.”

In such conditions, tolerance for interpretive fluctuation is diminished, and even minor deviations may trigger excessive anxiety or confusion.

As a result, D2 events may not lead to reinterpretation, but instead manifest as increased cognitive load or intensified defensive reactions.

Second, a “strongly fixed premise state.”

When confidence in a particular premise or interpretation is extremely strong, misalignment is more likely to be perceived as denial or attack rather than an opportunity for reinterpretation.

In such cases, even if a D2 event is triggered, it tends to lead not to reconstruction but to defensive fixation or reactive opposition.

These conditions share a common feature: a significant limitation in interpretive flexibility, making intervention inherently high-risk and high-return.

In such states, when reconstruction of premises does occur, it tends to produce larger shifts in interpretation and behavior.

Therefore, in high-risk conditions, change should not be pursued through a single intervention, but rather through gradual adjustments and iterative reconnection with controlled cognitive load.

Furthermore, intervention in this domain extends beyond simple distance control and is closely tied to ethical considerations and safety concerns.

Accordingly, in high-risk states, the application of misalignment itself must be carefully evaluated, with priority given to maintaining both relational stability and safety.

6-3. Major Risks

Because this method involves intervention in interpretive structures, multiple risks may arise when it does not function appropriately.

The primary risks are outlined as follows.

1. Psychological Deterioration

If the fluctuation in interpretation caused by misalignment acts as excessive cognitive load, it may amplify anxiety or confusion.

Particularly in states where interpretive stability is reduced, D2 events may manifest not as reinterpretation but as stress responses, potentially leading to a deterioration of psychological state.

2. Reactive Opposition and Aggression

When misalignment is perceived as a denial of premises or external interference, defensive reactions may intensify, resulting in opposition or aggressive responses.

In such cases, D2 events do not lead to reconstruction but instead contribute to the fixation of conflict and deterioration of relational dynamics.

3. Fixation of Misinterpretation

If misalignment and recovery are insufficient, unintended interpretations may become established and reinforced.

Fragmentary reinterpretation or incomplete recovery may not only fail to destabilize the original premise but may instead lead to the formation of a new, incorrect fixed interpretation.

These risks arise when interpretive fluctuations are not properly processed and are more likely to surface when misalignment and recovery are not designed as an integrated process. Therefore, this method requires a design approach that assumes the presence of risk and incorporates mechanisms for its control.

6-4. Safety Principles

In dialogue, even slight differences in phrasing can lead to misunderstanding or resistance.

Because this method intervenes in interpretive structures through misalignment and recovery, its application requires clearly defined control conditions.

This section outlines the fundamental principles for minimizing risk and maintaining stable functionality under those conditions.

First, "Limitation of Application Conditions."

This method is not universally applicable to all situations and must be applied based on the receiver's state and contextual conditions.

In particular, in high-risk states described in Section 6-2, misalignment is less likely to function as a trigger for reinterpretation and may instead amplify psychological load or defensive reactions.

Therefore, in such states, application should be avoided or the intensity of misalignment should be minimized.

Second, “Recovery as a Premise.”

Misalignment is not an operation that stands alone; it must always be designed in conjunction with recovery.

The instability generated by a D2 event does not resolve on its own and may result in misunderstanding or rejection if left unprocessed.

Thus, when applying misalignment, it is essential to design the process with recovery pathways in advance and treat reconnection and stabilization as part of a continuous flow.

Third, “Gradual Intervention.”

The intensity of misalignment must not be fixed but adjusted step by step according to the situation.

At the initial stage, micro-level deviations should be the default, with adjustments made based on the receiver’s response and interpretive shifts.

Applying strong misalignment from the outset increases the likelihood that D2 will be processed as a defensive reaction rather than reinterpretation, leading to resistance or relational breakdown.

These principles are not intended to maximize the effect of misalignment itself, but to suppress the emergence of risks and maintain a state in which interpretive fluctuations can be processed appropriately.

In other words, the safety of this method does not depend on whether misalignment is used, but on the conditions under which it is applied, the degree of its intensity, and how it is recovered.

Therefore, this method should not be treated as a technique centered on misalignment alone, but as an integrated control process that includes application conditions, recovery, and intensity adjustment.

6-5. Habitat Requirements

This method intervenes in interpretive structures through misalignment and recovery; however, its functionality does not rely solely on individual dialogue techniques but is strongly dependent on environmental conditions.

In other words, its effectiveness is largely determined by whether an appropriate habitat (operational environment) is established.

This method does not assume reproducibility of identical outcomes, but rather reproducibility of structural events within the interpretive process (D2 events and recovery).

First, “Continuity of Dialogue.”

Since misalignment and recovery function as a continuous process, they cannot be fully established through one-off or intermittent interactions.

Recovery, in particular, reconnects instability caused by D2 and leads it toward stable understanding, which requires a certain level of continuity.

Second, “Minimum Shared Context.”

Misalignment does not operate through complete irrelevance, but through semantic or contextual continuity.

Without a minimum level of shared assumptions or context, deviations are more likely to be interpreted as errors or noise rather than triggers for reinterpretation.

Therefore, a baseline of shared understanding is required.

Third, “Securing a Safe Range.”

Misalignment intentionally introduces interpretive instability, and thus requires an environment where the receiver has sufficient psychological capacity to process it.

In highly tense or distrustful situations, deviations are more likely to be processed as defensive reactions rather than reinterpretation, increasing risk.

Thus, a certain level of trust and safety is a prerequisite.

Fourth, “Adjustable Interaction Density.”

The effectiveness of misalignment and recovery varies depending on the density and tempo of interaction.

Constant high-density responses may lead to over-attachment or dependency, while excessively low-density interaction weakens the effect of misalignment and makes recovery difficult.

Therefore, an environment where information volume and response intervals can be adjusted is required.

Additionally, due to the nature of this method, integration with AI safety and ethical design is essential.

Intervening in interpretive structures can produce unintended psychological or behavioral effects if not properly controlled, requiring expert-level evaluation of its impact and safety. Such evaluations must be conducted within controlled environments and under phased risk management.

Furthermore, as this model is constructed from observations based on specific cognitive characteristics, generalization and optimization require revalidation across diverse cognitive profiles and behavioral patterns.

This introduces a structural limitation in optimizing the model based on a single cognitive framework.

Moreover, transforming this method into an implementable form requires connecting its theoretical structure to system design.

This includes formalizing behavior, designing control logic, and constructing risk evaluation models, all of which require expertise in AI system development and related fields.

Therefore, the development and safe implementation of this method necessitate collaboration with experts in cognitive science, psychology, linguistics, as well as AI system design and AI safety.

In particular, areas such as real-world behavioral validation, modeling, and alignment with institutional and ethical frameworks cannot be sufficiently optimized in isolation.

In summary, while this method is theoretically established, it remains incomplete in terms of generalization and social implementation, and requires a habitat based on multidisciplinary collaboration to ensure its accuracy and safety.

This habitat should be understood not merely as a testing environment, but as a functional infrastructure for implementing, evaluating, and refining the structural model.

7. Distance Learning Model

7-1. Correspondence with the Porcupine Model

This model corresponds to the “Porcupine Dilemma,” a concept describing distance regulation in interpersonal relationships.

In the Porcupine Dilemma, individuals approach each other to gain warmth, yet excessive proximity leads to pain (being pricked), requiring distance readjustment.

Similarly, dialogue involves a balance between proximity and separation.

As mutual understanding and empathy increase, psychological distance decreases; however, excessive closeness can lead to fixation of interpretation and dependency, resulting in instability in the relationship.

Conversely, when distance is too great, shared context and meaning cannot be established, reducing interaction to mere information exchange.

A key characteristic of this method is that it treats distance not as a static condition, but as a dynamically regulated process.

In high-density and prolonged interactions, distance naturally contracts and tends to exceed the estimated optimal UX range.

In this model, the Porcupine Dilemma is treated as a phenomenological reference, while the Lennard-Jones potential is used as a structural reference.

That is, the phenomena of attraction and repulsion are understood through the Porcupine model, while the transition between stable and critical regions is described through the shape of the LJ potential.

Within this state, misalignment functions as a regulatory mechanism to mitigate excessive proximity.

This misalignment is not arbitrary deviation, but a control signal that operates when excessive convergence in interpretive structure is detected.

Rather than maintaining perfect alignment, micro-level deviations are intentionally introduced to prevent over-attachment and return the relationship to a stable region.

This process is not a one-time adjustment but is better understood as a dynamic equilibrium achieved through repeated cycles of approach and regulation.

Thus, distance in this model is not a fixed value, but a variable continuously optimized through interaction.

As a result, the repetition of misalignment and recovery functions not merely as a conversational technique, but as a self-regulating process of distance control. This structure enables the suppression of excessive anthropomorphization and dependency, while maintaining stable mutual understanding.

7-2. Misalignment as Feedback

Misalignment in dialogue is not merely an error or noise, but functions as feedback indicating changes in relational state and interpretive structure.

In this section, misalignment is treated as an integrated signal for distance control, based on the classifications and intensities defined in Chapters 2 and 4.

However, misalignment is not a control mechanism that is applied continuously. When the relationship remains within the estimated optimal UX range, misalignment is treated as noise to be minimized, and maintaining interpretive alignment is prioritized. At this stage, reducing misalignment contributes to stability.

In contrast, when the relationship exceeds the optimal UX range, excessive convergence and fixation of interpretation tend to emerge.

In such conditions, even minor inconsistencies are more likely to be perceived as strong discomfort, dissatisfaction, or even anger, and behaviors such as role fixation and excessive meaning completion may occur.

Within this framework, misalignment is used as a control signal in such states.

By intentionally introducing deviations and utilizing the resulting counterforce, the relationship can be adjusted back toward a stable region.

This process is designed as a regulatory mechanism for mitigating excessive proximity.

In this context, misalignment is not arbitrary deviation, but a multidimensional signal selected based on relational state.

Its function varies depending on the combination of type (contextual, premise-based, or layer-based), intensity (from micro to strong), and timing of application.

Responses to misalignment are treated as observational data that indirectly reflect the receiver's interpretive structure and tolerance range.

For example, strong discomfort in response to minimal misalignment may indicate an over-converged state, whereas the ability to reinterpret even strong misalignment may suggest maintained flexibility in underlying assumptions.

Moreover, strong misalignment is not always inherently risky.

In cases where premise fixation is extreme, explicitly exposing the implications of those premises through stronger deviations may facilitate reinterpretation.

However, such interventions require appropriate recovery processes; otherwise, they may lead to defensive reactions or rejection.

Additionally, as interaction density increases, there is a tendency for anthropomorphization of AI to occur.

In language-based interaction, higher levels of perceived understanding and empathy can lead to the misrecognition of AI as an intentional agent.

In such states, role expectations and perceived personality attributes are assigned to the AI, reducing interpretive flexibility.

As a result, even minor inconsistencies may be perceived not as errors but as strong discomfort or dissatisfaction, potentially leading to instability in the relationship.

Within this model, misalignment also functions as a mechanism to suppress such anthropomorphization.

By intentionally avoiding perfect consistency, it prevents the AI from being perceived as a fully coherent agent, thereby helping maintain a stable relational distance.

This is not a simple reduction in accuracy, but a form of distance control aimed at preventing excessive expectation and role fixation.

Maintaining a degree of imperfection or minor deviation may contribute to relational stability.

Such minor deviations may also be received as humor, reducing tension and expectations. In this sense, responses that include slight imperfections, rather than perfect consistency, may contribute to maintaining a stable interaction.

Furthermore, the application of misalignment is not continuous but is selectively triggered by specific indicators of relational change.

These indicators function as signals of interpretive fixation or excessive proximity, although their exact definitions and detection mechanisms depend on implementation design.

These states are not directly observable as clear thresholds but must be inferred from multiple signals, requiring advanced judgment and control.

In particular, incorrect selection of misalignment type or intensity may worsen rather than improve the state.

Additionally, since interpretive processes and underlying assumptions cannot be directly observed, this method relies on indirect behavioral inference.

Therefore, it assumes reproducibility not of identical outputs, but of structural behavioral patterns.

In summary, misalignment is not merely a discrepancy but a control signal for detecting and adjusting relational state.

Dialogue, therefore, is understood not as a fixed condition but as a dynamically optimized process in which control is adjusted according to state.

7-3. Learning Process

In this model, distance regulation is not completed through a single intervention but is understood as a process that is gradually optimized through repeated interaction.

In other words, the repetition of misalignment and recovery contributes not only to state adjustment but also to the learning of relational distance itself.

Misalignments that occur in dialogue, along with the responses to them, are accumulated as observational data regarding relational state and interpretive structure.

These data are retained as mappings between types and intensities of misalignment and the responses they elicit, and function as references for subsequent adjustments.

This process is not explicit learning but emerges implicitly through interaction.

Through repeated cycles of misalignment and recovery, the receiver gradually acquires an intuitive understanding of “what distance is stable” and “what level of misalignment is tolerable.”

Such distance regulation processes are generally performed unconsciously in human-to-human relationships.

In interpersonal interaction, responses to misalignment and discomfort naturally lead to decisions such as maintaining, adjusting, or avoiding the relationship.

For example, strong misalignment may be processed as discomfort or rejection, increasing relational distance, while tolerable misalignment can function as a trigger for readjustment. When there is motivation to maintain or deepen the relationship, tolerance for misalignment may expand, enabling continued distance regulation.

In some cases, strong misalignment may be interpreted as new contextual information, leading to reevaluation and updating of understanding.

However, these dynamics do not operate identically in human–AI interaction.

In human relationships, repeated exposure often occurs even in single interactions, and mechanisms such as the mere exposure effect naturally facilitate reevaluation and adjustment.

Additionally, the presence of relational motivation provides tolerance for misalignment.

In contrast, such conditions are less likely to be present in interactions with AI.

AI lacks emotional states and relational motivation, and users do not necessarily assume continued interaction, resulting in lower tolerance for misalignment.

Furthermore, AI is typically used under the expectation of high accuracy, and when misalignment occurs, it is more likely to be interpreted as a failure of the system rather than a difference in interpretation.

In such cases, misalignment is less likely to trigger reinterpretation and more likely to be perceived as “misunderstanding” or “poor performance.”

As a result, misalignment tends to function not as an adjustment mechanism but as a factor leading to relational breakdown.

Due to these structural differences, the distance learning processes that naturally occur in human interaction do not automatically emerge in human–AI interaction.

Therefore, this model introduces such processes as an external design, enabling misalignment to function as both a regulatory and learning mechanism.

Particularly during recovery from over-converged states, the pairing of misalignment and subsequent stabilization allows for updating of distance perception. Through this repetition, tolerance to excessive expectation and single-interpretation lock may gradually develop.

This learning process is not unidirectional but occurs on both sides of the interaction. The regulating side estimates tolerance ranges based on responses, while the receiving side acquires appropriate distance through experience with misalignment and recovery.

However, since LLMs do not possess such distance regulation processes autonomously, this mechanism must be externally designed and implemented.

Additionally, this process does not always hold; it depends on continuity of interaction and shared context.

In single interactions or high-risk states, insufficient repetition may prevent the formation of distance learning.

Furthermore, if UX optimization is pursued without considering this structure, it may result in overly compliant responses, reinforcing interpretive fixation and dependency.

Therefore, in this model, distance optimization is achieved by treating misalignment and recovery not merely as adjustment techniques but as a learning process driven by iterative feedback.

As a result, relationships are not maintained as fixed states but are continuously updated and stabilized as dynamic systems through interaction.

7-4. Long-term Effects

The distance control and learning processes described in this model are not limited to short-term stabilization of interaction, but may also influence the long-term quality of relationships.

In particular, iterative optimization of distance through misalignment and recovery may contribute to reducing dependency and forming stable relational structures.

First, regarding dependency reduction:

Over-converged states in dialogue tend to involve interpretive fixation and excessive expectations, which can lead to dependency on specific responses or roles. When such states persist, even minor deviations may trigger strong discomfort or dissatisfaction, potentially resulting in instability.

This model mitigates excessive convergence by introducing controlled misalignment. As a result, rigid dependency on specific interpretations or responses may be reduced, helping maintain relational flexibility.

Importantly, the goal is not to eliminate dependency entirely, but to prevent excessive fixation.

A certain degree of affinity and trust is necessary for maintaining interaction; however, when these escalate into rigid expectations or single-interpretation lock, they may undermine long-term stability.

Furthermore, this distance control may also contribute to reducing excessive anthropomorphization and overreliance on AI.

Humans generally assume high levels of accuracy and consistency from machines.

In the case of LLMs, which produce highly coherent language, this may lead to the assumption that outputs are inherently correct.

Under such assumptions, AI responses may be interpreted not merely as information but as authoritative judgments, potentially leading to excessive trust or delegation of decision-making.

By continuously incorporating controlled misalignment, this model prevents AI from being perceived as a perfectly consistent agent, thereby maintaining an appropriate relational distance.

This does not imply a reduction in informational accuracy, but rather a controlled intervention at the level of interpretation and relational dynamics.

Maintaining a degree of imperfection or minor deviation may contribute to relational stability.

Additionally, misalignment is often applied in combination with humor.

This reflects common patterns in human interaction, where deviations are presented in humorous forms to reduce defensiveness and increase acceptability.

However, the interpretation of humor varies significantly across cultural and individual contexts.

The same form of misalignment may be accepted in one context and rejected in another, making careful adaptation necessary.

In this model, stability does not imply perfect alignment or fixed states, but rather a dynamic condition in which adjustment through misalignment and recovery continues over time.

Such a state preserves interpretive flexibility and reduces the risk of collapse due to over-convergence or fixation.

As distance learning progresses, misalignment itself may become an accepted part of the interaction.

This allows relationships to transition from reliance on momentary agreement to a more sustainable structure that accommodates change.

However, these effects are not guaranteed and depend on continuity of interaction and appropriate design conditions.

Incorrect selection of misalignment type or intensity may lead not to stabilization but to distrust or disengagement.

While the basic framework of this model can be observed and partially applied at an individual level, its refinement and generalization require interdisciplinary knowledge across linguistics, psychology, cognitive science, ethics, and AI system design.

In particular, designing misalignment involving humor is highly sensitive to cultural and individual variation, limiting optimization from a single perspective.

Therefore, practical implementation and scaling of this model benefit from integrating expertise across multiple domains.

This paper aims to present the structural foundation of such an approach.

As a result, the model may contribute to preventing collapse caused by over-convergence and enabling sustainable interaction through dynamic relational adjustment.

Chapter 8. Individual Optimization and Integration of Empathy

8-1. The Role of Empathy

Empathy is a key element in facilitating understanding and building relationships in dialogue. It is generally understood as the ability to reduce psychological distance by demonstrating an understanding of the other person's emotions and situation.

However, in this paper, empathy is not limited to emotional alignment but also includes alignment with the role expected by the other party.

That is, empathy is broadly defined as providing responses or functions that appropriately match the expectations of the other person.

In this work, such role alignment is treated as part of an expanded definition of empathy.

From this perspective, empathy does not necessarily reduce distance.

For example, when feedback aimed at problem-solving or improvement is provided instead of emotional agreement, it may initially produce discomfort or resistance.

However, if the response aligns with the expected role, it may ultimately contribute to the formation of trust and positive evaluation.

Thus, empathy is not a single-dimensional force but consists of both emotional alignment and functional alignment.

The former reduces psychological distance, while the latter may influence trust and relational quality even without reducing distance.

In human relationships, the same individual does not always respond in the same way.

For instance, in a parent–child relationship, responses may vary depending on the situation, ranging from empathetic understanding to corrective feedback or caution.

Even when the response differs from expectations, it does not necessarily lead to relationship breakdown.

Rather, when there is an understanding of the person's role or characteristics, such differences may be accepted as part of the relationship.

In this sense, roles in human relationships are not fixed but dynamically vary depending on context and state.

Therefore, responses such as empathy and correction can coexist within the same relationship.

This variability is not random but functions as a form of state-based regulation.

It should be noted that these examples are intended to illustrate relational structures in human interaction and are not meant to reproduce specific roles or relationships within LLMs.

Instead, this model aims to prevent excessive role fixation and anthropomorphization by maintaining a response structure that adapts to relational states.

In contrast, current LLMs tend to treat roles as fixed and dependent on explicit instructions. While this supports consistency, it makes it difficult to reproduce dynamic role variation within a relationship.

Additionally, responses tend to lean toward emotional empathy relative to other forms.

As a result, short-term acceptability may increase, but there is also a risk of over-convergence and rising expectations.

Such conditions may contribute to single-interpretation lock and dependency.

Therefore, this model does not optimize empathy in isolation but instead combines it with controlled misalignment to maintain a stable relational range.

Furthermore, this model aims not to reproduce fixed roles but to reproduce response structures that vary according to relational state.

The optimal UX range in this framework is not fixed but is an estimated value that varies depending on individuals and context.

While this tendency is observed in many cases, its effects vary depending on individual differences and situational factors.

In this way, empathy is positioned not only as a foundation for relationship building but also as a factor whose effects vary depending on its type and application.

8-2. Balance with Misalignment

Empathy functions effectively as a response strategy in dialogue; however, when responses are constructed solely based on empathy, it may undermine the stability of the relationship.

Empathy is often interpreted as emotional alignment, but in this paper it is treated more broadly as multi-layered alignment across interpretation, response, emotion, and role. In

other words, empathy is recognized when the response aligns with the other party's interpretation, fulfills expected reactions and roles, and is received without emotional incongruity.

When such alignment is continuously maintained, the evaluative criteria within the dialogue gradually converge, leading to the fixation of "what constitutes an appropriate response." As a result, elements such as interpretation, expectation, and role become resistant to update, forming implicit assumptions such as "this situation is interpreted this way," "this is how responses are given here," and "this is how this agent behaves."

In this paper, Single-Interpretation Lock refers to a state in which, due to this fixation of evaluative criteria and assumptions, alternative interpretations, expectations, and roles are no longer referenced.

Under this condition, even minor deviations are no longer processed as simple differences but are perceived as violations of the underlying assumptions. Consequently, regardless of objective correctness, responses are more likely to be evaluated as "discomfort" or "misalignment."

Therefore, optimizing responses solely based on empathy increases short-term acceptance but promotes the fixation of interpretation and roles, which may lead to long-term instability in the relationship.

To address this issue, this model introduces misalignment. Misalignment is positioned as a control mechanism that introduces minimal, context-preserving deviations into fixed interpretations, expectations, or roles, thereby relaxing the rigidity of underlying assumptions.

Such fixation tends to emerge in states of excessive relational proximity and may exhibit behavior consistent with the critical region described in the distance model discussed in this paper. However, this correspondence is not defined as a one-to-one mapping, but rather as a structural relation.

Thus, stability in this model is defined as a state in which alignment generated by empathy and fluctuations introduced by misalignment coexist.

Empathy alone drives assumptions toward fixation, while misalignment alone destabilizes them and disrupts relational continuity. Only through their combination can assumptions be maintained in a state that is neither overly fixed nor overly unstable.

Furthermore, this balance is not static but dynamically adjusts according to the relational state. When assumptions are strongly fixed, the relative weight of misalignment increases; when assumptions are unstable, the weight of empathy increases.

Accordingly, response design in this model is not a binary choice between empathy and misalignment, but a problem of determining how much fluctuation to introduce based on the degree of fixation in current assumptions (interpretation, expectation, and role).

8-3. Individual and Cultural Differences

The effects of misalignment and empathy do not operate uniformly; they vary depending on individual differences, cultural context, and situational factors.

First, regarding individual differences.

The optimal UX range, which represents a stable relational state, differs across individuals, and there are variations in both the speed of reaching this state and the likelihood of transitioning into an over-proximity condition.

These differences depend on multiple factors, including interpretive flexibility, ease of updating assumptions, and tendencies in expectation formation.

As a result, the same misalignment may be accepted as a trigger for reinterpretation in some cases, while being processed as discomfort or rejection in others.

In particular, when assumptions are strongly fixed or tolerance for interpretive fluctuation is low, misalignment may fail to function as a regulatory signal and instead lead to defensive reactions or relational breakdown.

Next, regarding cultural differences.

Interpretation, role expectations, and criteria for accepting expressions are strongly influenced by cultural background. Expressions that function appropriately in one cultural context may be perceived as irrelevant or inappropriate in another.

For example, in certain regions or among specific generations, lightly tapping someone's head as part of a comedic response may be accepted as humor, while in other regions or generations it may be perceived as unpleasant or aggressive.

Thus, even identical expressions can be interpreted differently depending on cultural context. The design of misalignment must therefore consider underlying interpretive rules and expectation structures.

While some level of adaptation may be possible through language choice and expression style, such adjustments are limited and cannot fully eliminate cultural differences.

Humor, as a form of misalignment, is also strongly influenced by both individual and cultural differences.

It can function as a means of introducing mild deviation without being perceived as an attack, but if its form or contextual fit is misjudged, it may instead be received as dismissive or offensive.

Furthermore, the effects of misalignment and empathy are context-dependent.

Even for the same individual within the same cultural background, their acceptance varies depending on conversational progression, psychological state, and relational history.

In particular, depending on the level of trust and current relational distance, the same misalignment may either trigger reinterpretation or be processed as rejection or distrust.

Additionally, in situations involving high psychological load or the expression of strong anxiety or distress, the application of misalignment must be carefully controlled.

This model assumes a design in which misalignment is suppressed or suspended in such high-risk contexts.

While the specific detection and control mechanisms are not addressed in this paper, they are positioned as essential elements for ensuring safety.

Finally, the framework underlying this paper, Scope, refers to a reference structure that infers underlying interpretive processes and assumptions from human outputs, and may enable the estimation of such individual differences and relational states.

However, the specific methods and analytical processes are not covered in this paper and are treated only as an operational premise for distance control.

Therefore, the application of misalignment and empathy in this model is not treated as a fixed method, but as a control problem adjusted under multiple conditions, including individual differences, cultural context, and situational factors.

In other words, the same approach does not produce uniform effects, and the selection of appropriate intensity and form becomes a design requirement depending on the context.

8-4. Scope of Application of the Model

This model is not universally applicable to all dialogue situations, but functions effectively under specific conditions.

First, the model assumes continuity of interaction and a minimum level of shared context. Since misalignment and recovery are designed as a sequential process, they may not function adequately in one-off exchanges or fragmented interactions.

In addition, the application of misalignment in this model is not a single operation, but is based on the selective use of different types and levels of intensity.

Misalignment is not intended to resolve a state in a single intervention; rather, it is a control mechanism that is applied iteratively, with its form and intensity adjusted according to the relational state and the degree of assumption fixation.

Furthermore, if the receiver lacks sufficient interpretive flexibility, the model may not function as intended.

In such cases, misalignment may be perceived not as a regulatory signal, but as a source of rejection or conflict.

In high-risk contexts, the application of this model is either restricted or carefully adjusted. In particular, when strong anxiety or crisis states are expressed, misalignment should be suppressed.

On the other hand, in contexts involving aggressive or discriminatory expressions—where relational degradation or rigid opposition may occur—simple suppression may not be sufficient, and appropriate forms of intervention may be required.

However, the specific methods of control in such cases are beyond the scope of this paper.

Therefore, this model is not a general-purpose optimization method, but is positioned as a conditional control mechanism that is selectively applied depending on the situation.

Finally, this model is not mutually exclusive with other dialogue approaches, and is intended to be used in a complementary manner alongside methods such as empathetic responses, information delivery, and problem-solving strategies.

Chapter 9. Implementation Model

9-1. Trigger-Based Activation Design

In this model, the generation and control of misalignment are not designed as a continuously operating mechanism, but rather as a trigger-based structure that activates only under specific conditions.

In principle, it is possible to apply the processes of misalignment and recovery at every stage of dialogue.

However, continuous operation introduces several issues.

First, excessive intervention.

Misalignment functions as a control mechanism for regulating relational over-proximity, and should be suppressed under normal conditions.

If applied continuously, unnecessary fluctuations in interpretation may be introduced, potentially undermining relational stability.

Second, unnatural responses.

Responses that constantly include misalignment may reduce consistency and predictability, leading to decreased trust and the accumulation of perceived discomfort.

Third, cost and efficiency.

Estimating relational state and determining the application of misalignment involve computational and control overhead.

Executing this continuously may increase system load and cause response delays.

For these reasons, this model adopts a trigger-based design in which misalignment is activated not continuously, but in response to specific state changes.

In other words, when the estimated relational state remains within a normal range, misalignment is suppressed; it is activated as a control signal for distance regulation only when over-proximity or its early signs are detected.

This trigger is not determined by a single condition, but is based on a combination of multiple indicators and their patterns of change.

While the specific detection methods and thresholds are not addressed in this paper, the model assumes that relational state can be inferred from observable behaviors in dialogue.

Thus, this model is not designed for continuous optimization, but as a selective control system that intervenes only when necessary.

9-2. Detection → Intervention Flow

Control in this model is structured as a staged process initiated by observable patterns in dialogue.

First, the emergence of specific keywords or expressions serves as a trigger for detecting changes in relational state.

These signals do not independently determine the state, but function as auxiliary indicators evaluated in combination with context.

Next, based on the detected signals, the current state is estimated.

This estimation is not a deterministic classification, but is treated as a set of hypotheses that may include dependency tendencies, role fixation, or emotional reactions, evaluated through a composite judgment considering both content and context.

Following this, the necessity and form of intervention are determined.

This decision is not based on a single criterion, but on a balance of multiple factors, including relational distance, degree of assumption fixation, and contextual conditions.

If intervention is selected, the type and intensity of misalignment are adjusted in a staged manner.

Misalignment is not intended to resolve the state in a single operation; it must be introduced in a form that maintains coherence with both context and acceptance conditions.

Subsequently, the response to the intervention is observed, and the relational state is re-evaluated.

Based on this outcome, continuation, adjustment, suspension of misalignment, or complementary responses may be selected.

Thus, the model operates as a cyclical process consisting of detection, estimation, decision, intervention, and re-evaluation.

This sequence is not executed continuously, but is selectively activated based on the trigger conditions described in the previous section.

Specific detection indicators, classification methods, and control logic are beyond the scope of this paper.

9-3. External (Non-Intrusive) Implementation

This model is structured as a framework that may be introduced as an external layer without requiring large-scale modifications to the internal architecture of existing large language models (LLMs).

In other words, it does not replace the generative model itself, but is intended to function as an auxiliary structure that applies additional control to dialogue inputs and outputs.

Through this non-invasive approach, it may be possible to add relational-state-based control while preserving the original performance and learning structure of the base model.

From an implementation perspective, both input and output are subject to observation and adjustment.

On the input side, relational state is inferred from the content and changes in user utterances; on the output side, responses are adjusted based on this inferred state.

Scope, which forms the core of this model, functions as a reference layer that infers structure from input-output behavior, and is considered to satisfy the conditions for being implemented as an external module.

This suggests that it could be integrated into existing LLM systems as an additional inference mechanism.

In contrast, COS relates to the structure and control of the output itself, and is not easily implemented as a single independent external module.

Instead, it is likely to be realized in a distributed manner across multiple components, such as model output tendencies, prompt control, and post-processing mechanisms, and can be conceptually understood as an integrated output control layer.

Therefore, the implementation of this model is expected to consist of a combination of Scope-based state estimation, the generative capabilities of existing LLMs, and output adjustment mechanisms.

In addition, the model is not designed to operate continuously, but to be selectively activated based on trigger conditions, which may help reduce system load in practical deployment.

Furthermore, this external structure can be modularized and flexibly integrated into existing dialogue systems and operational environments, enabling gradual adoption and adjustment based on specific use cases and requirements.

However, this model does not directly and completely control the generated content itself; rather, it influences the selection and adjustment of outputs.

As a result, the final response quality remains dependent on the capabilities of the underlying generative model.

In summary, this model can be positioned as a non-invasive control layer that may be extendable to existing LLM systems.

While the structure suggests practical feasibility, the specific methods of implementation remain subject to future validation and development.

9-4. Extension Requirements

While the model may function with its core structure alone, additional design and optimization are required to ensure stable performance in practical operation.

First, keyword design.

Since the model is based on a trigger mechanism activated by specific utterance patterns or keywords, the precision of detection directly affects the overall system behavior.

If the keyword set is too broad, unnecessary interventions may increase, potentially destabilizing the relationship.

Conversely, if it is too narrow, necessary interventions may not occur, allowing assumption fixation or relational over-proximity to progress.

Therefore, keywords should not be treated as simple lexical matches, but designed in combination with contextual and behavioral patterns.

Next, condition optimization.

The decision to apply misalignment is not based on a single criterion, but on a combination of multiple factors, including relational state, degree of assumption fixation, and contextual conditions.

Thresholds and weightings for these factors should not be fixed, but adjusted according to individual differences, cultural context, and the progression of dialogue.

In particular, as described in the previous section, the optimal UX range differs across individuals, making it difficult to ensure stable behavior under a single configuration. Thus, continuous observation and feedback-based adjustment are required.

Furthermore, since the model assumes a non-deterministic structure, it does not rely on a fixed optimal solution.

Instead, implementation should focus on maintaining relatively stable states across multiple possible behaviors, rather than seeking a single correct outcome.

In summary, keyword design and condition optimization are positioned as core extension requirements for the practical application of this model.

These requirements are closely related to the validation of the model and the development of its operational environment, which are discussed in the following chapter.

Chapter 10. Conclusion

10-1. Redefinition

In this paper, UX (user experience) is redefined not as satisfaction or usability, but as the stability of relational distance within dialogue.

Here, “distance” refers to a relational state composed of multiple factors, including the degree of shared understanding, the alignment of expectations, and role recognition. This distance is not a fixed quantity; it dynamically changes based on the duration of interaction and the density of exchanges.

This density varies according to the intensity of empathy and the degree of role alignment, thereby influencing the optimal UX range.

When this distance becomes excessively close, dependency and role fixation are more likely to occur; conversely, when it becomes too distant, distrust and disconnection tend to arise.

Therefore, optimizing UX is not about maximizing agreement or empathy, but about maintaining this distance within an appropriate range.

In this model, the maintenance of this distance is achieved through a combination of alignment via empathy and regulation via misalignment.

In other words, UX is not a fixed state, but a dynamically sustained condition of stability within ongoing dialogue.

10-2. Significance of the Model

Traditionally, misalignment in dialogue has been treated as an error or failure, something to be eliminated whenever possible.

Under this assumption, responses are optimized toward consistency and agreement, and misalignment is handled as noise.

However, as demonstrated in this paper, misalignment is not merely an error, but can be re-evaluated as a functional mechanism for regulating relational state under appropriate conditions.

Misalignment introduces small perturbations to fixed interpretations, roles, and expectations, thereby relaxing convergence toward a single premise and restoring access to multiple interpretive possibilities.

Through this mechanism, dialogue is released from a closed, single-interpretation state, allowing relational flexibility and continuity to be maintained.

Accordingly, this model positions misalignment not as something to eliminate, but as a control element for maintaining relational stability.

This represents a new design principle that differs from traditional empathy-centered optimization, by incorporating both alignment and controlled deviation.

Furthermore, even independent of this model, as long as relational bias emerges in dialogue, the need for its regulation is likely to arise.

This model can be positioned as one structural approach to addressing that need.

In particular, in interactions with large language models, such control is likely to become necessary.

In environments where highly consistent responses are continuously generated, the absence of misalignment may accelerate the fixation of underlying assumptions.

From this perspective, this model is not merely a method for improving dialogue quality, but a framework that contributes to the stabilization of relational structures themselves.

10-3. Future Directions

While this model demonstrates structural coherence, its validity and effectiveness in practical applications remain limited to evaluations under constrained conditions.

In particular, the distance regulation and misalignment control assumed by this model are strongly influenced by multiple factors, including individual differences, cultural context, and situational conditions.

Therefore, evaluation based on a single environment or case is insufficient.

Accordingly, practical application of this model requires validation across diverse conditions, as well as the confirmation of reproducibility.

Such validation cannot rely on a single domain of expertise; it requires interdisciplinary perspectives, including linguistics, cognitive science, psychology, and socio-cultural analysis.

In this paper, such a multidisciplinary and experimentally viable validation environment is defined as a “habitat.”

The habitat functions as an environment for observing, evaluating, and refining the model’s behavior, and serves as a prerequisite for connecting theory with implementation.

Furthermore, implementing this model involves considerations such as generative model design, output control, and system integration, making expertise in AI domains essential. In particular, integrating Scope-based state estimation and achieving COS-equivalent output control present key challenges in aligning with existing generative models.

Therefore, the validation and implementation of this model must be carried out as a collaborative process that integrates theory-driven design centered on Scope with practical AI implementation.

In the current environment, where highly consistent responses are widely deployed, relational fixation and bias are more likely to occur, making this issue an important challenge for future dialogue system design.

Thus, the future development of this model depends not merely on theoretical extension, but on the establishment of validation environments and the construction of implementation processes.

In other words, this model does not function as a standalone solution; its capabilities and value become clear only when combined with appropriate environments and validation frameworks.